



KUMARAGURU

COLLEGE OF TECHNOLOGY

NAAC (National Assessment and
Accreditation Council)

GRADE
A++

24ADT015 FINANCE, ECONOMICS AND MARKETING

Unit -1 Managerial Economics & Demand Estimation

Prepared By :
Mohan Kumar R S
Asst. Prof.
Dept of Mech Engg.

Contents



**Managerial goals vs
Shareholder value;**



**Law of demand &
supply;**



**Elasticity (price,
income, cross);**



**Marginal utility &
consumer surplus;**



**Exceptions to the
law of demand;**



**Quantitative &
qualitative demand-
forecasting
techniques**



**(time-series, causal
models, Delphi, big-
data tools);**



**Introduction to
behavioural
economics for
managers.**

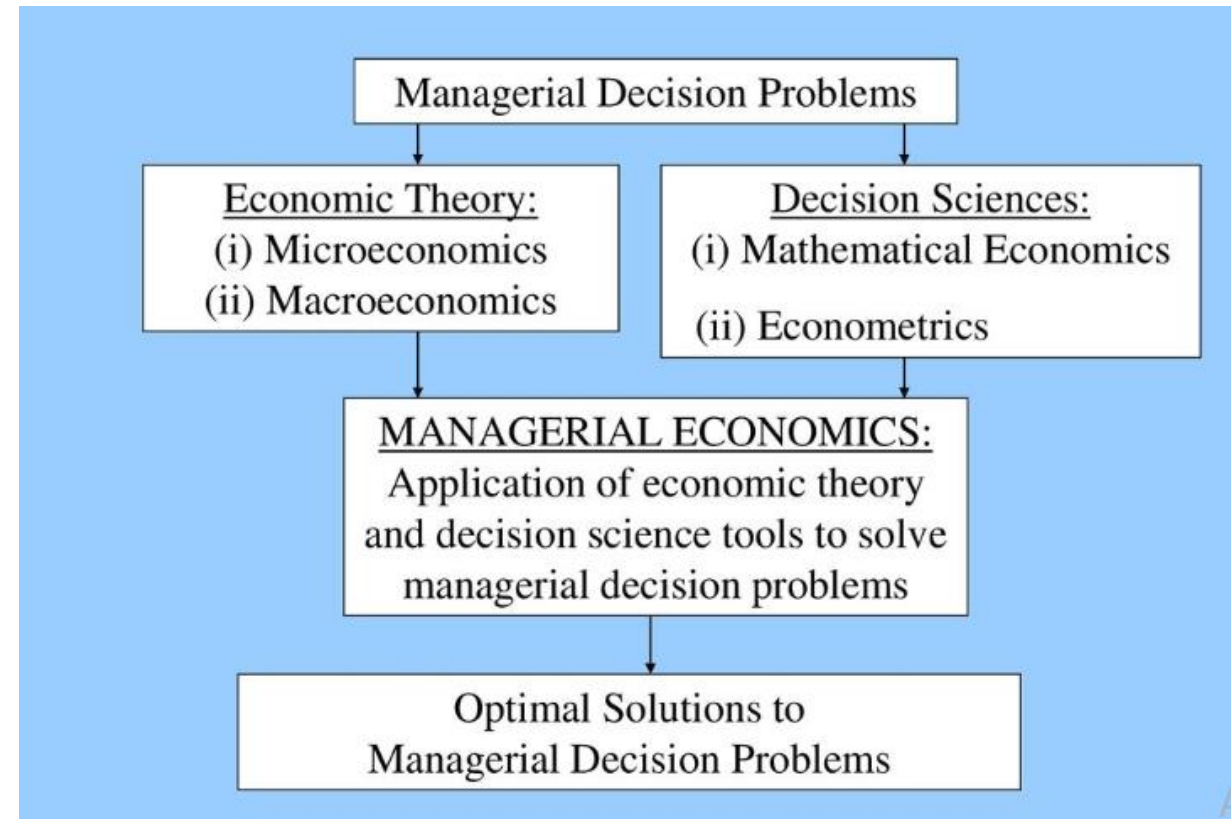
What is Managerial Economics?



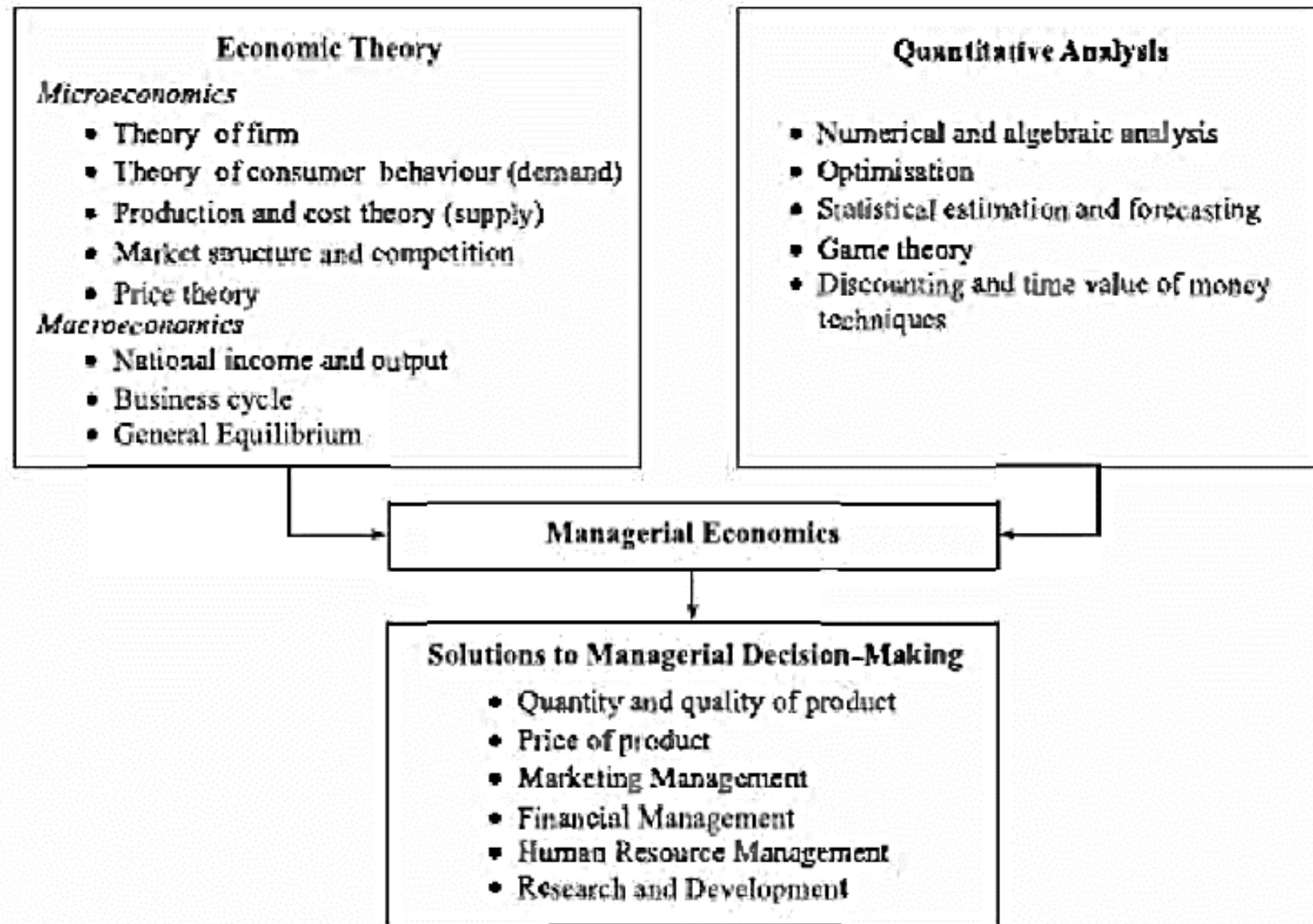
Economics

- Economics is a social science, since it deals with the society as a whole and human behavior in particular, and studies the production, distribution and consumption of goods and services.

Relation of Managerial Economics With Decision Sciences



Relation Of Managerial Economics With Decision Sciences



- **Microeconomics** - looks at the smaller picture of the economy and is the study of the behavior of small economic units, such as that of an individual consumer, a seller (or a producer, or a firm), or a product.
- **Macroeconomics** - branch of economic analysis that deals with the study of aggregate demand and aggregate supply, GDP, GNP, national income, national capital formation, employment, inflation etc.





Managerial Economics

- **Managerial economics** is finding the most efficient way of allocating scarce organizational resources reaching stated objectives of firm

(ie., Profitability, Market Share, etc.,)

- Managerial economics applies economic theory and methods in business and administrative decision making.

Managerial goals vs Shareholder value

Managerial Goals



Employee
Satisfaction



Market
Share



Social
Responsibility



Social
Responsibility



Market
Share



Market
Share



vs

Shareholder Value



ROI
Share



Profitability
ROI



Stock
Price



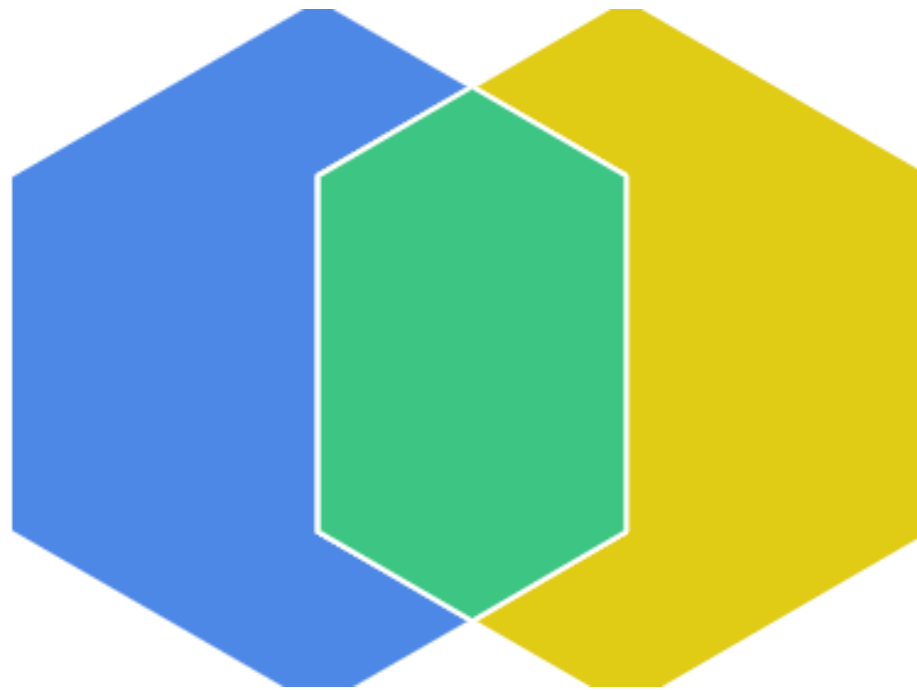
Stock
ROI



Stock
Price

Managerial Goals

Internal organizational
objectives



Shareholder Value

Financial expectations and
returns

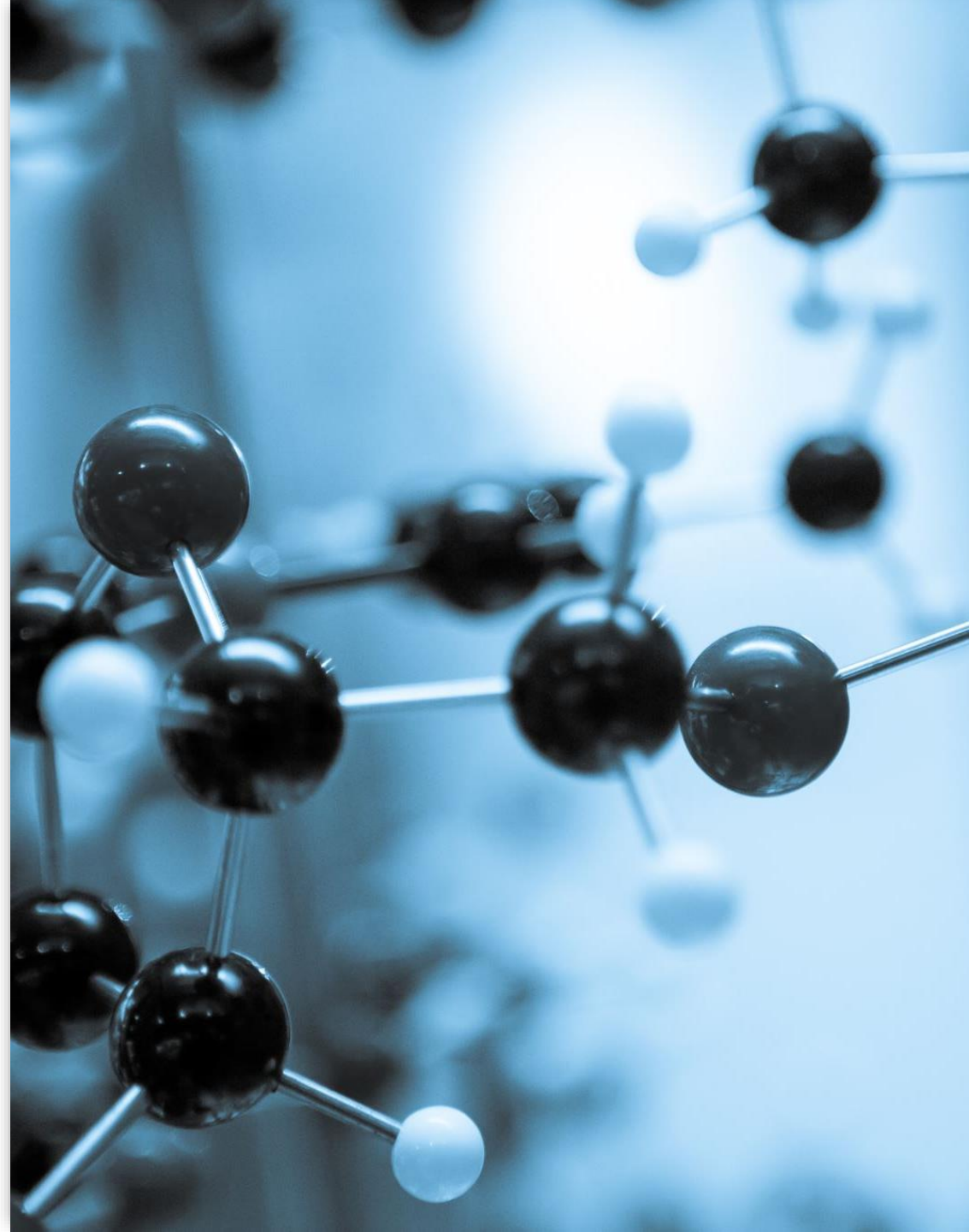
-
- **Managerial goals vs Shareholder value**
 - In any organization there's often a tension between what management wants to achieve (**Managerial Goals**) and what shareholders expect (**Shareholder Value**).
 - As future IT professionals, you'll **build systems, analyze data, and contribute to decisions** that directly impact both managerial objectives and shareholder wealth.

Relation of Managerial Economics with Decision Sciences

- If Q^d is the quantity demanded of a commodity by a firm that depends on factors like P (price of the commodity),
- P_y (price of a related commodity),
- Y (income of consumer),
- t (tastes of consumers),

then the demand function can be written as:

$$Q^d = f(P, P_y, Y, t)$$



LAW OF DEMAND



- **The Law of Demand**

- Demand refers to the **quantity of a good or service that consumers are willing and able to purchase at various prices** during a specific period.

- **Statement:** As the price of a good or service increases, the quantity demanded decreases, and vice versa, assuming all other factors remain constant.

Law of Supply

The Law of Supply states that an increase in price leads to an increase in supply and vice versa.



- **The Law of Supply**

- Supply refers to the **quantity of a good or service that producers are willing and able to offer for sale at various prices** during a specific period.

- **Statement:** As the price of a good or service increases, the quantity supplied increases, and vice versa.“

EXPLORING ELASTICITY IN CONSUMER BEHAVIOR

Understanding how price and income changes affect demand and supply



Understanding **elasticity** concepts

Elasticity measures responsiveness to price changes.

Types of **elasticity**

Includes price, income, and cross-price **elasticity**.

Impact on **pricing strategies**

Managers tailor strategies based on **elasticity** insights.

Role in **consumer behavior**

Helps in predicting how consumers react to changes.

Consumer Surplus: Measuring Economic Benefit

Defining Consumer Surplus

Consumer surplus quantifies the monetary gain a consumer receives when the price they are willing to pay for a good or service exceeds its actual market price. It represents the **extra value** or satisfaction gained beyond the cost incurred.

Example: Your Morning Coffee

Imagine you're willing to pay £10 for your favourite artisanal coffee. However, the coffee shop only charges £7. Your consumer surplus in this transaction is £3. This £3 is the benefit you receive from the market price being lower than your personal valuation.



Marginal Utility: Satisfaction Increments

The Law of Diminishing Returns

Marginal utility refers to the additional satisfaction or benefit a consumer derives from consuming **one more unit** of a good or service. This concept is crucial for understanding demand curves and how consumers allocate their budgets.

First Slice of Pizza

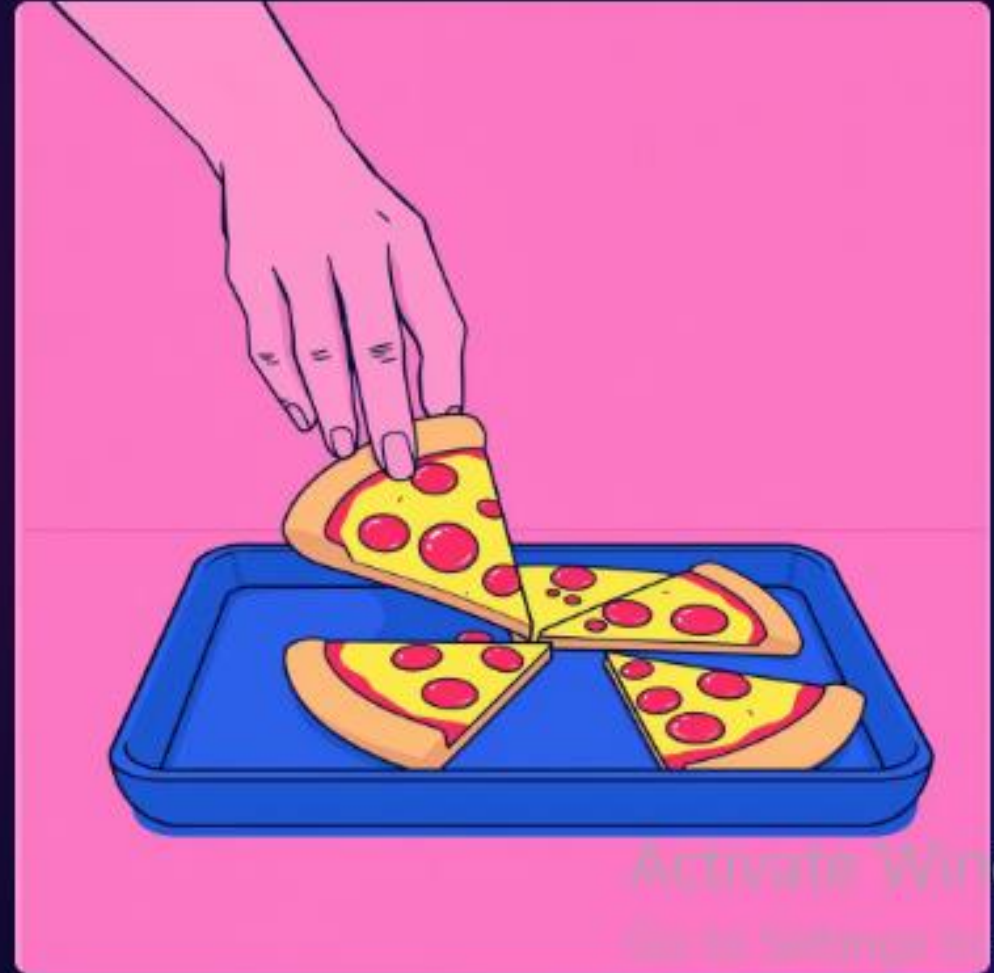


The first slice of pizza when you're hungry brings immense satisfaction – high marginal utility.

Subsequent Slices



As you eat more slices, the additional satisfaction from each subsequent slice typically decreases, illustrating the law of diminishing marginal utility.



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UNDERSTANDING EXCEPTIONS TO THE LAW OF DEMAND

Explore how Giffen and Veblen goods challenge traditional economic principles

Giffen Goods Dynamics

Giffen goods illustrate a unique scenario where demand increases as prices rise, primarily due to the **income effect** outweighing the **substitution effect**. This challenges conventional demand

Veblen Goods Appeal

Veblen goods represent luxury items whose demand rises with price increases, as higher prices enhance their desirability as **status symbols**. This reflects the influence of **social factors** on consumer behavior.

Implications for Demand Forecasting

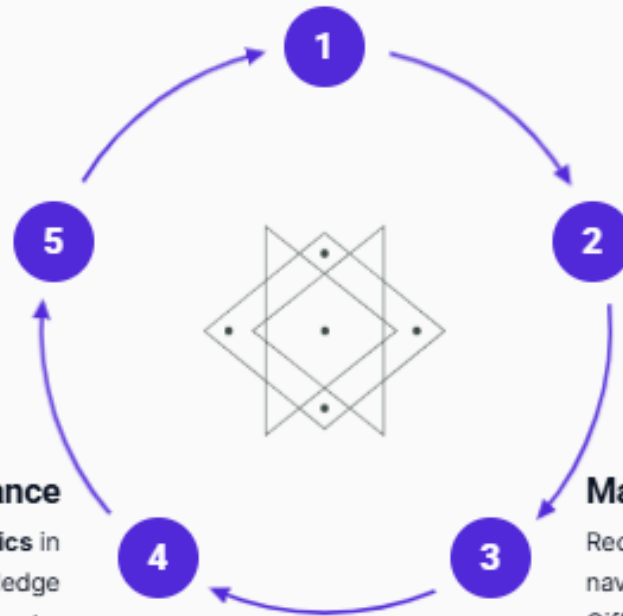
Understanding exceptions to the law of demand is crucial for **effective demand forecasting**. It enables more accurate predictions and tailored marketing strategies, especially for **luxury** and **necessity** goods.

Behavioral Economics Relevance

These exceptions underscore the importance of **behavioral economics** in understanding consumer choices. Managers can leverage this knowledge to predict and influence demand more effectively in various segments.

Market Condition Insights

Recognizing these exceptions allows managers to better understand and navigate **unique market conditions**. By adapting strategies in response to Giffen and Veblen goods, businesses can enhance their market positioning.



EXPLORING QUANTITATIVE DEMAND FORECASTING TECHNIQUES

Understanding statistical methods to enhance inventory management and production alignment for better market performance.



UNDERSTANDING QUALITATIVE DEMAND FORECASTING TECHNIQUES

Exploring expert-driven methods for predicting future market demands effectively

1 Qualitative demand forecasting relies on expert insights

This approach prioritizes the **judgment** of experts and their understanding of market dynamics over numerical data, making it essential in forecasting for new products.

2 The Delphi method fosters expert consensus

By gathering opinions from a panel of experts, the **Delphi method** aims to achieve a consensus on future demand, enabling more accurate predictions in uncertain markets.

3 Ideal for new products and limited data

Qualitative techniques are particularly beneficial for **new product forecasting** or scenarios where historical data is scarce, allowing managers to gauge potential market interest effectively.

4 Complementing quantitative methods

Managers can use qualitative forecasting to enhance **quantitative methods**, providing a more comprehensive understanding of market trends and demands.

5 Incorporates market research findings

Qualitative approaches often incorporate insights from **market research**, helping to validate assumptions and refine forecasts based on real-world observations.

6 Useful in volatile market conditions

In rapidly changing environments, qualitative techniques can adapt more swiftly than quantitative models, making them vital for timely decision-making.

7 Enhances strategic planning

By understanding potential demand through qualitative techniques, companies can improve their **strategic planning** and resource allocation, aligning with market needs.

What Are Time Series and Why Do They Matter in Economics?

Understanding Time Series Data

Time series data are sequential observations of economic variables, such as monthly retail sales or quarterly GDP, recorded at regular intervals. This chronological order is crucial for revealing underlying patterns.

Key Components Explained

Every time series typically comprises three key components: a long-term **trend** (direction), **seasonality** (calendar-related patterns), and **irregular fluctuations** (unpredictable variations).

Significance in Economics

Time series analysis is essential for understanding complex economic dynamics, rigorously testing economic theories, and making precise forecasts of future economic conditions, aiding critical decision-making.

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Core Principles & Econometric Foundations



Causality Defined

For true causality, the cause must precede the effect and not be confounded by hidden variables or external factors.



Structural Models

Structural causal models utilise systems of equations to link endogenous (dependent) and exogenous (independent) economic variables.



Econometric Techniques

Techniques like instrumental variables and difference-in-differences are crucial for isolating and identifying causal effects.



Controlling Confounders

Actively controlling for confounding variables is paramount to accurately isolating true causal relationships.

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The Path Forward: Building & Applying Causal Models

Define Theory & Model

Start by clearly defining your economic theory, then specify a robust structural model based on this foundation.

Estimate with Data

Proceed to estimate your model using relevant economic data, employing appropriate econometric techniques.

Identify Assumptions

Crucially, identify and articulate all necessary assumptions to ensure the validity and reliability of your model.

Validate & Apply

Finally, validate your findings against real-world cases and apply the model to derive actionable economic insights.

Mastery of causal econometrics empowers economists to make better decisions and generate robust economic insights. Embrace causal thinking to transform raw data into invaluable, actionable economic knowledge.

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How the Delphi Method Works: Step-by-Step

1

Step 1: Define & Select

Clearly define the economic issue and select a diverse panel of experts relevant to the topic.

2

Step 2: Initial Forecasts

In Round 1, experts provide open-ended forecasts and opinions anonymously through questionnaires.

3

Step 3: Feedback & Summary

A facilitator summarises responses and shares anonymised feedback, including statistical analysis of opinions.

4

Step 4: Refinement Rounds

Subsequent rounds allow experts to refine views based on group feedback, narrowing differences in forecasts.

Why Big Data Changes Economics

1

Beyond Traditional Econometrics

Traditional econometrics handle small, structured data well. Big data, with its immense volume, velocity, and variety, demands new analytical approaches.

2

New Data Sources

Smart sensors, transaction logs, and social media generate terabytes daily, enabling granular economic analysis previously impossible.

3

Computational Challenges

Handling millions of predictors and complex, real-time relationships requires advanced computational methods and scalable infrastructure.

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The Future: Big Data Empowering Evidence-Based Economics

1

Enhanced Forecasting & Patterns

Big data analytics improves forecasting and uncovers hidden patterns, supporting data-driven policies.

2

Targeted Interventions

Harvard's Opportunity Insights, for example, uses big data to study economic mobility and design effective interventions.

3

New Skills & Paradigms

Economists must adopt new analytic paradigms and coding skills to harness big data's full potential for social good.

The era of big data tools marks a transformative leap in understanding and shaping economic outcomes, fostering a more informed and impactful approach to policy.

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HARNESSING BIG DATA FOR ACCURATE DEMAND FORECASTING

Leveraging advanced analytics to enhance predictions and respond to market changes effectively

Revolutionizing Forecasting

Big data tools have transformed the landscape of demand forecasting by analyzing extensive datasets from multiple sources, allowing for enhanced accuracy in predictions.

Enhanced Customer Satisfaction

By tailoring offerings based on accurate forecasts, businesses can significantly enhance **customer satisfaction**, ultimately driving loyalty and sales growth.



Identifying Patterns

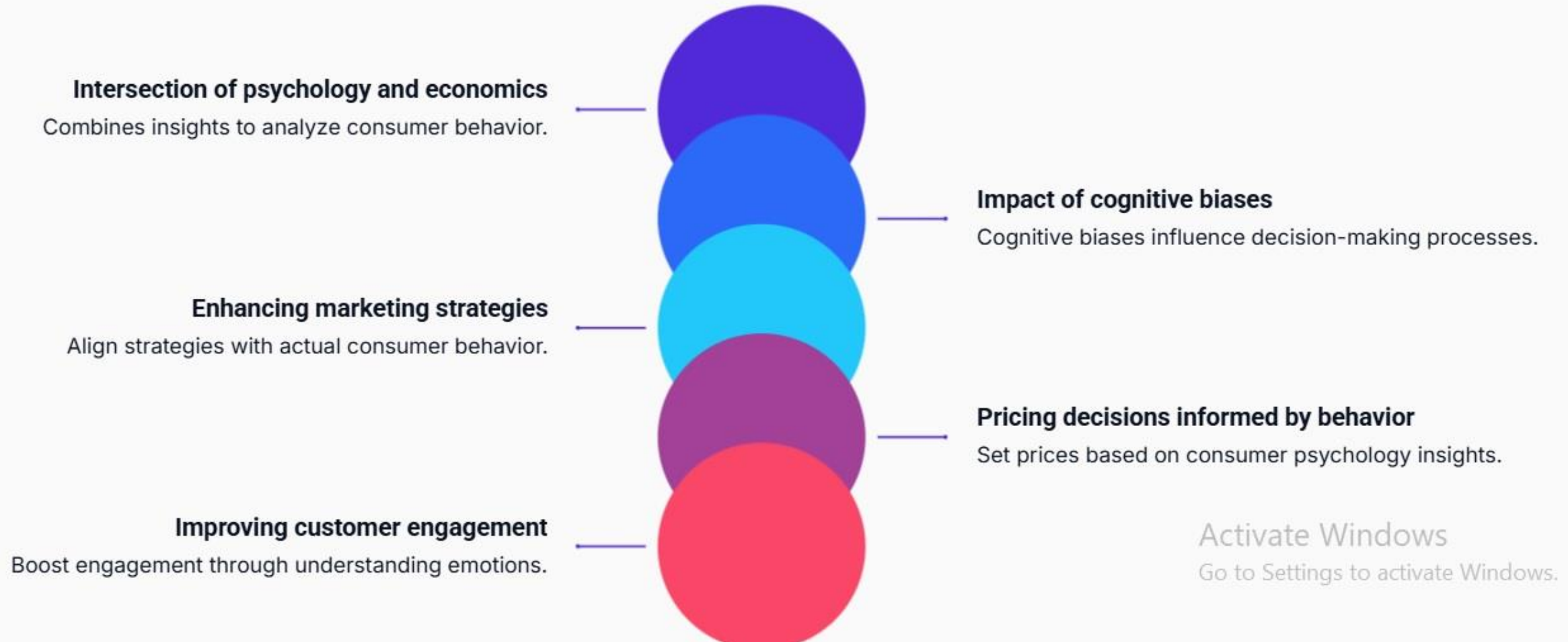
These tools uncover hidden **patterns** and **trends** that traditional forecasting methods might miss, enabling businesses to make more informed decisions.

Real-Time Adjustments

With **real-time data analytics**, managers can dynamically adjust strategies, swiftly responding to market fluctuations and improving overall operational efficiency.

UNDERSTANDING BEHAVIORAL ECONOMICS FOR MANAGERS

Exploring consumer behavior through psychology and economics principles



BEHAVIORAL ECONOMICS: RETAIL SUCCESS STORY

Exploring the impact of pricing strategies on consumer behavior and sales

30% Sales Increase A significant boost in sales achieved through behavioral strategies.	Loss Aversion Consumer Influence Utilizing psychological insights to shape purchasing behavior.
Dynamic Pricing Pricing Strategy Flexible pricing methods to adapt to market demands.	Promotional Offers Engagement Tactic Framing offers to enhance consumer attraction.
Behavioral Insights Strategy Foundation Understanding consumer psychology to inform business choices.	Retail Chain Business Context The case study focuses on a specific retail entity.

INTEGRATING ECONOMIC THEORY WITH MANAGEMENT PRACTICE

Understanding demand, elasticity, and behavioral economics for effective management strategies

1

Incorporating economic theories into management practices.

Managers can enhance decision-making by applying **economic principles** to anticipate market trends and consumer behavior, leading to improved outcomes.

2

Understanding the law of demand and supply is crucial.

By grasping the dynamics of **demand and supply**, managers can make informed decisions on product availability and pricing strategies, optimizing their market position.

3

Utilizing elasticity to refine pricing strategies.

Managers can leverage **price elasticity** and **income elasticity** insights to set competitive prices that maximize revenue while remaining attractive to consumers.

4

Recognizing marginal utility and consumer surplus.

Understanding how **marginal utility** affects consumer choices helps managers tailor offerings to maximize **consumer surplus**, enhancing customer satisfaction and loyalty.

5

Identifying exceptions to the law of demand.

Awareness of **exceptions to the law of demand**, such as Veblen and Giffen goods, enables managers to craft nuanced strategies that cater to unique consumer behavior.

6

Employing quantitative and qualitative demand forecasting techniques.

Using methods like **time-series analysis** and **Delphi techniques** allows managers to predict future demand accurately, informing inventory and production decisions.

7

Integrating behavioral economics insights into management.

Applying principles from **behavioral economics** helps managers understand psychological factors influencing consumer decisions, leading to better marketing and engagement strategies.

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NAVIGATING FUTURE TRENDS IN DEMAND FORECASTING AND BEHAVIORAL ECONOMICS

Exploring advancements in data analytics, AI, and consumer insights for effective decision-making

ADVANCEMENTS IN DATA ANALYTICS

Data analytics tools will significantly improve the accuracy of demand forecasts, allowing for better market predictions.

IMPACT OF ARTIFICIAL INTELLIGENCE

AI-driven insights will empower managers to make informed decisions based on predictive analytics, improving market responsiveness.

UNDERSTANDING CONSUMER BEHAVIOR

Deep insights into consumer behavior will help managers tailor their strategies to meet the evolving demands of their customer base.

DECISION-MAKING ENHANCEMENT

Utilizing advanced forecasting tools will enhance the overall decision-making process, leading to better business outcomes.

ROLE OF MACHINE LEARNING

Machine learning algorithms will analyze vast datasets to identify patterns in consumer behavior, enhancing forecasting accuracy.

NEED FOR INNOVATION IN FORECASTING

Businesses must innovate their forecasting approaches to adapt to rapidly changing consumer preferences and market conditions.

IMPORTANCE OF STAYING UPDATED

Managers should continuously update their knowledge on trends in demand forecasting to leverage new opportunities effectively.

LET'S LEVERAGE ECONOMIC INSIGHTS FOR STRATEGIC ADVANTAGE AND GROWTH.

Embrace the principles of demand, supply, elasticity, and behavioral economics to optimize your strategies. Apply these insights to improve marketing efforts, boost overall performance, and navigate the competitive landscape effectively.

